

TITLE PAGE

Afatinib is an oral targeted anticancer drug used mainly for certain types of lung cancer.

Afatinib, the world's first irreversible ErbB family (containing four different cancer cell epidermal growth factor receptors, including EGFR, HER2, ErbB3, and ErbB4) inhibitor, is a second-generation oral epidermal growth factor receptor tyrosine kinase inhibitor (EGFR-TKI). It can be used as a first-line treatment for locally advanced or metastatic non-small-cell lung cancer (NSCLC) with an EGFR-sensitive mutation or for patients with locally advanced or metastatic squamous lung cancer whose disease progresses during or after platinum-containing chemotherapy. Currently, with the use of third-generation EGFR-TKIs, afatinib is no longer clinically indicated as the first choice for patients with NSCLC who have EGFR-sensitive mutations

Created by - Pooja Gursale

Modified By – Vihari Gollapally

Reviewed by – Amol Dhawale

TABLE OF CONTENTS**SUMMARY**

Afatinib is an oral cancer medication used to treat non-small cell lung cancer (NSCLC) with specific EGFR gene mutations by irreversibly blocking EGFR and Her2 tyrosine kinases. See section 6.

What Afatinib Is

Afatinib, sold under the brand name Gilotrif, is a tyrosine kinase inhibitor that targets the epidermal growth factor receptor (EGFR) and human epidermal growth factor receptor 2 (Her2). It works by

irreversibly binding to these receptors, preventing cancer cells from growing and dividing uncontrollably, which is particularly effective in NSCLC cases with certain EGFR mutations

Afatinib is primarily prescribed for:

Non-small cell lung cancer (NSCLC) with EGFR-positive mutations, including exon 18 mutations (e.g., G719X) and deletion 19-positive tumors

Patients who have metastatic NSCLC or have previously received platinum-based chemotherapy but did not respond well See Figure A.

It is being investigated for other cancers driven by EGFR or Her2, such as certain breast cancers

Afatinib is included in the World Health Organization's List of Essential Medicines due to its clinical importance refer section 5 – Section 6.

1 Basic Information

- **Class:** Tyrosine kinase inhibitor (TKI)
- **Brand name:** *Gilotrif*

- **Route:** Oral (tablet) See Tankere, P 2022.

2 Mechanism of Action

Afatinib irreversibly inhibits the ErbB family receptors, including see Chiu, CH 2015

- EGFR (Epidermal Growth Factor Receptor)
- HER2 (ErbB2)

This blocks signaling pathways responsible for:

- Cancer cell growth
- Proliferation
- Survival

3 Indications (Uses)

Primarily used in Non-Small Cell Lung Cancer (NSCLC) See Ullas, B 2022

3.1 First-line treatment

- a. For metastatic NSCLC with EGFR mutations (e.g., exon 19 deletion, exon 21 L858R)
See Table 1- Table 3.

3.2 Squamous NSCLC:

- b. After progression on platinum-based chemotherapy See Figure 1- Figure 2.

4 Introduction

In current history, great strides have been made in the treatment of advanced non-small-cell lung cancer (NSCLC), and there is a consensus that epidermal growth factor receptor tyrosine kinase inhibitor (EGFR-TKI) therapy is preferred for patients who are EGFR-positive. In non-Asian groups, EGFR mutations account for 10% to 15% of NSCLC cases, while the incidence of such mutations rises to 40% in Asian populations. The majority of EGFR mutations are exon 19 deletion mutations and exon 21 L858R point mutations, collectively referred to as common EGFR mutations, which are responsible for 80% to 90% of the total EGFR mutations observed. Patients carrying common EGFR mutations are the most beneficial population for EGFR-TKI molecular targeted therapy. The remaining approximately 10% of EGFR mutations are considered rare [1]. These variations contain exon 20 insertions and/or point mutations (such as S768I), compound mutations (such as S768I+G719X), uncommon mutations in exon 19, and less frequent exon 21 mutations (such as L861Q). Among them are the missense mutations G719X, S768I, and L861Q, respectively, which are common mutations among uncommon mutations and are known as significant rare mutations. See Section 2.

It is noteworthy that, unlike common mutations, these uncommon mutations exhibit varying degrees of sensitivity to EGFR-TKIs.

In a retrospective investigation conducted in China, the clinical traits and therapeutic outcomes of individuals with NSCLC with common and uncommon EGFR mutations were evaluated. According to the study, patients with uncommon mutations who received first-line platinum-based chemotherapy or first-generation EGFR-TKIs seemed to have a lower objective response rate (ORR) and median progression-free survival (PFS) than clients with common mutations. Section 1 (ORR: 23.3% vs. 51.8%, $p = 0.003$; median PFS: 7.1 vs. 10.9 months, $p = 0.001$) [5]. Gefitinib and erlotinib, first-generation EGFR-TKIs, were investigated for their efficacy for the treatment of lung adenocarcinomas with the G719X, L861Q, and S768I mutations in a multicenter study from Taiwan. The outcomes were compared to those of patients with classical EGFR mutations. The results suggested that after receiving first-generation EGFR-TKIs, compared to patients with classical mutations, individuals with rare mutations had a significantly poorer ORR (41.6% vs. 66.5%, $p < 0.001$), median PFS (7.7 vs. 11.4 months, $p < 0.001$), and median overall survival (OS) (17.2 vs. 27.8 months, $p < 0.001$) [6]. A comprehensive worldwide retrospective study called UpSwinG was conducted to assess the effectiveness of NSCLC patients with uncommon EGFR gene mutations. Afatinib patients had a median time to treatment failure (TTF) of 11.3 months (95% CI: 8.5–14.9), which was higher than the 8.8-month (95% CI: 6.4–10.7) TTF shown in patients undergoing first-generation EGFR-TKIs [7]. According to the research mentioned above, chemotherapy or first-generation EGFR-TKIs are less profitable for treating NSCLC patients with uncommon EGFR mutations than they are for treating patients with common EGFR mutations. Compared to first-generation EGFR-TKIs and chemotherapy, afatinib may be a superior option for patients with uncommon EGFR mutations. Refer to Appendix 1 – Appendix 2.

In 2018, the Food and Drug Administration (FDA) of the United States authorized afatinib as a first-line therapy for individuals with the uncommon EGFR mutations G719X/L861Q/S768I. In light of this, the current paper aims to offer a thorough analysis of the effectiveness of afatinib in the therapy of NSCLC patients with various uncommon EGFR mutations mentioned in Appendix 1 and Table 2, which can help medical professionals decide what the most effective strategy for individuals with such mutations is.

5 Anti-Tumor Mechanism

Members of the ErbB family include the human epidermal growth factor receptor 2 (HER2), the epidermal growth factor receptor (EGFR), ErbB3, and ErbB4, which are all potentially inhibited by the medication afatinib. As an irreversible blocker, it binds covalently to the tyrosine kinase domain of these receptors, leading to their inactivation and preventing the downstream signaling pathways involved in cell growth and survival. Afatinib's distinctive anticancer therapeutic benefit—its irreversible binding to ErbB family receptors—offers higher potential to suppress the proliferation of a wide class of tumor cells and greater efficacy than the first-generation EGFR-TKIs. Afatinib works in NSCLC with common EGFR mutations through this pharmacological mechanism, and some individuals with rare EGFR mutations are also susceptible to it.

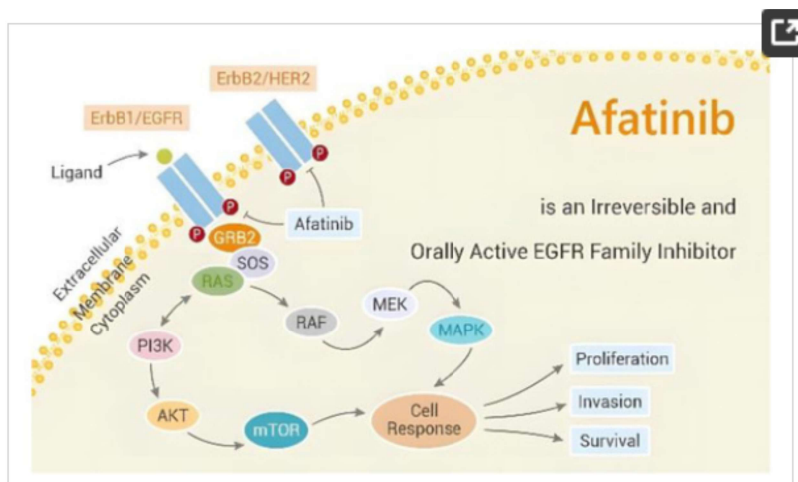


Figure A. Anti-tumor mechanism of afatinib (picture from the Network of Cancer Research).

6 References

Tankere, P.; Boidot, R.; Bonniaud, P.; Zouak, A.; Foucher, P.; Milliere, A.; Bertaut, A.; Favier, L.; Lagrange, A.; Ghiringhelli, F.; et al. Uncommon EGFR mutations in lung carcinoma: Features and treatment outcomes in a retrospective French cohort. *J. Thorac. Dis.* **2022**, *14*, 2034–2044.

Xu, H.; Yang, G.; Liu, R.; Yang, Y.; Li, W.; Li, J.; Hao, X.; Xing, P.; Wang, Y. EGFR uncommon alterations in advanced non-small cell lung cancer and structural insights into sensitivity to diverse tyrosine kinase inhibitors. *Front. Pharmacol.* **2022**, *13*, 976731

Ullas, B.; Shrinidhi, N.; Mansi, S.; Narayan, S.; Parveen, J.; Surender, D.; Joslia, J.T.; Anurag, M. All EGFR mutations are (not) created equal: Focus on uncommon EGFR mutations. *J. Cancer Res. Clin. Oncol.* **2022**, *149*, 1541–1549.

Shi, J.; Yang, H.; Jiang, T.; Li, X.; Zhao, C.; Zhang, L.; Zhao, S.; Liu, X.; Jia, Y.; Wang, Y.; et al. Uncommon EGFR mutations in a cohort of Chinese NSCLC patients and outcomes of first-line EGFR-TKIs and platinum-based chemotherapy. *Chin. J. Cancer Res.* **2017**, *29*, 543–552

Chiu, C.H.; Yang, C.T.; Shih, J.Y.; Huang, M.S.; Su, W.C.; Lai, R.S.; Wang, C.C.; Hsiao, S.H.; Lin, Y.C.; Ho, C.L.; et al. Epidermal Growth Factor Receptor Tyrosine Kinase Inhibitor Treatment Response in Advanced Lung Adenocarcinomas with G719X/L861Q/S768I Mutations. *J. Thorac. Oncol.* **2015**, *10*, 793–799.

7 Tables

Table 1. Afatinib's efficiency in treating NSCLC patients who had mutations in exon 18

Study	<i>n</i>	Mutation	Tumor Staging	Therapy Line	Response	PFS (mos) (95% CI)	OS (mos) (95% CI)
Hao Y., 2022 [18]	11	E709A/G/K, E709-T710delinsD	IV	first or above	N/A	13.5	17.1
Iwamoto Y., 2019 [19]	1	E709-T710delinsD	IV	sixth	PR	7	N.R.
Ibrahim U., 2017 [20]	1	E709-T710delinsD	IV	first	PR	N.R.	N.R.
An N., 2019 [21]	1	E709-T710delinsD	IIIA	second	PR	11	>21
Wei Y., 2021 [22]	1	E709-T710delinsD	IIB	first	PR	23	N.R.

Legend: NSCLC—non-small-cell lung cancer; N.R.—not reported; N/A—not applicable; PR—partial response; PFS—progression-free survival; OS—overall survival; mos—months.

Table 2. Afatinib's efficiency in treating NSCLC patients who had uncommon mutations in exon 19.

Study	n	Mutation	Tumor Staging	Therapy Line	ORR (%)	Response	PFS (mos) (95% CI)	OS (mos) (95% CI)
Liang S.K., 2019 [25]	5	L747S or L747P	IV	first	80	N/A	11.97 (6.37–17.57)	16.62 (3.05–30.19)
Shan B.B., 2022 [24]	1	I740_K745dup	IV	second	0	SD	13.4	21.3
Zhang L., 2022 [26]	1	E746_L747delinsIP	IV	fourth	100	PR	>6.5	N.R.
Kanbe M., 2022 [27]	1	L747V and 21 L858R	IV	sixth	100	PR	12	N.R.
Fang Y.F., 2021 [28]	1	T751_I759delinsS	IV	second	100	PR	9	N.R.

Legend: NSCLC—non-small-cell lung cancer; N.R.—not reported; N/A—not applicable; PR—partial response; SD—stable disease; ORR—objective response rate; PFS—progression-free survival; OS—overall survival; mos—months.

ZENSAR

CONFIDENTIAL

2034326535_00

Table 3. Afatinib's efficiency in treating NSCLC patients who had fully defined exon 20 insertion mutations.

Study	<i>n</i>	Mutation	ORR (%)	TTF (mos) (95% CI)
Yang J.C., 2022 [15]	42	Overall	33	9.1 (7.4–14.2)
	4	A763_Y764insFQEA; A763_V765dup	50	39.0 (8.2–39.0)
	5	A767_S768insSVA; _V769dup/ASV; insASVD	0	3.7 (1.0–36.0)
	3	D770_N771insGL/SVD	0	3.8 (3.0–20.1)
	9	H773_R776insYNPY; _V774dup/insH; dup	0	24.0 (6.1–NE)
	2	M766delinsMATL; insASV	100	12.9 (11.6–14.2)
	7	N771_H773dup; _772insPHGH; delinsKG; _P772insGY	71	10.0 (5.2–NE)
	5	S768_D770dup	0	8.5 (NE–NE)
	7	V769_770INSV; _D770insASV/GVV	75	8.0 (1.2–14.3)

Legend: NSCLC—non-small-cell lung cancer; ORR—objective response rate; TTF—time to treatment failure; mos—months.

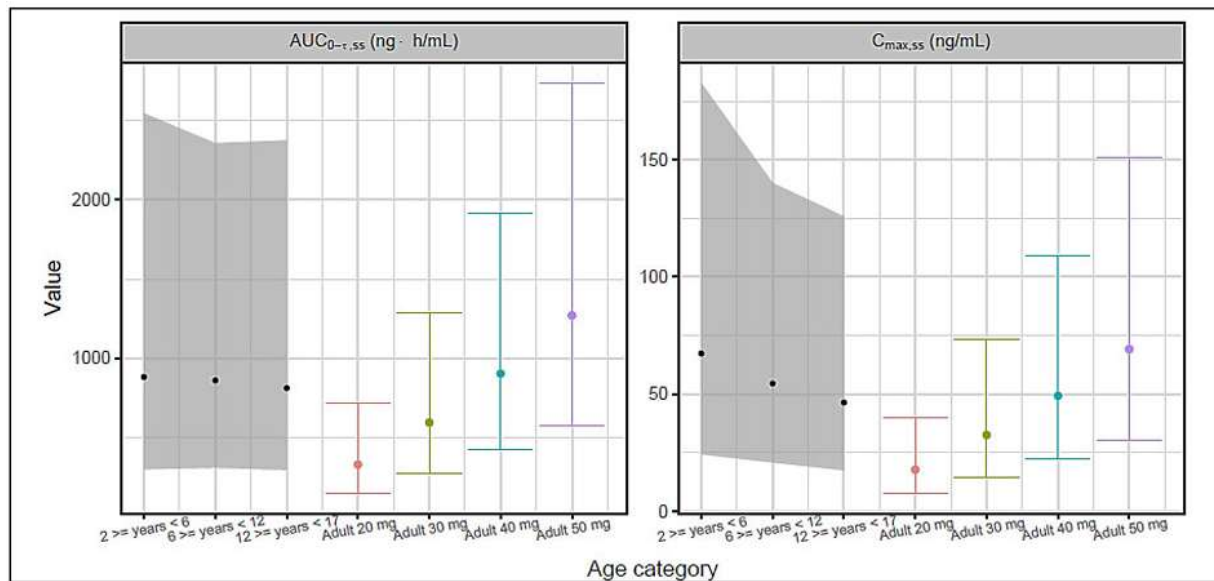
ZENSAR

CONFIDENTIAL

2034326535_00

8 Figures

Figure 1. Simulated afatinib AUC_{0-T,ss} and C_{max,ss} in the pediatric population versus adult values



Note: The point represent the 50th percentile and the shaded area is bounded by the 5th and 95th percentiles of the simulated data. The error bars represent the 5th and 95th percentiles of the adult data. 1000 pediatric subjects per age category were simulated.

ZENSAR

CONFIDENTIAL

2034326535_00

Figure 2. Comparison of individual and gMean of AUC_{0-24,ss} values of afatinib in pediatrics at 18mg/m² QD and adults at 40mg QD. (source: PopPK report - c31792800-01)

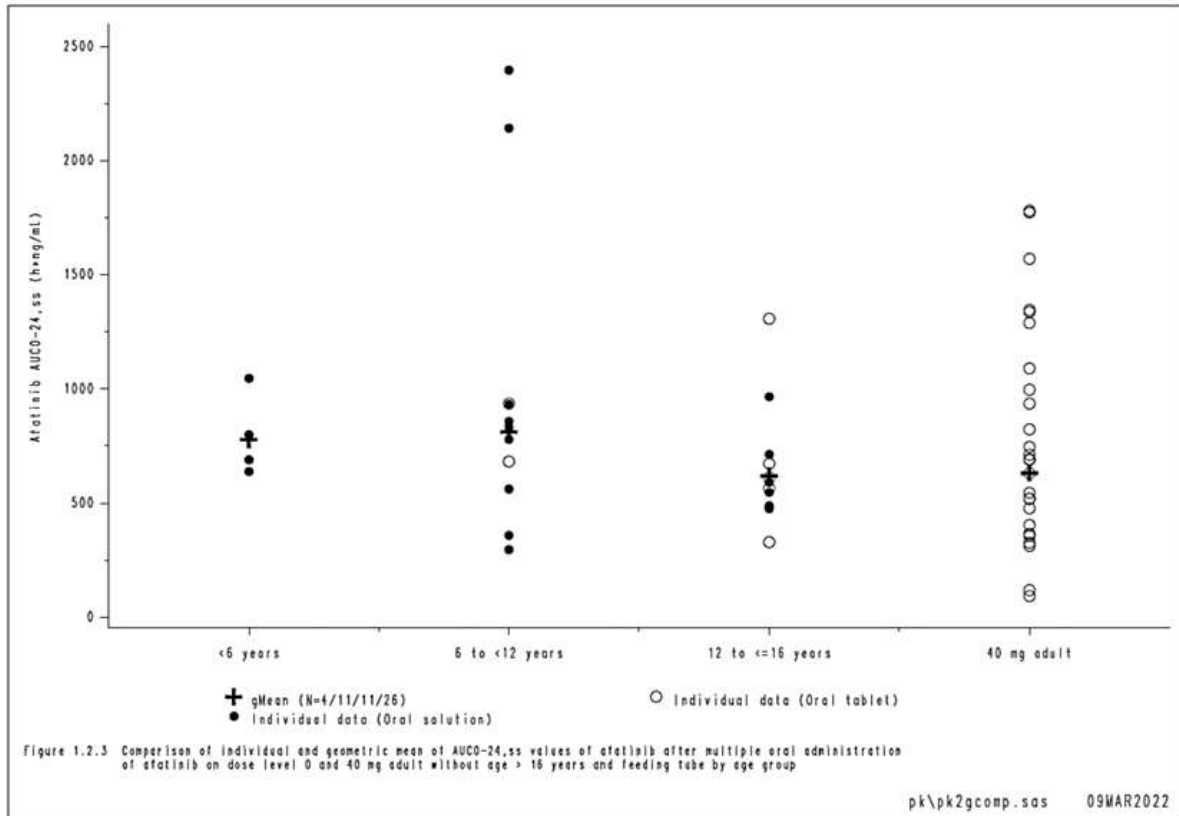
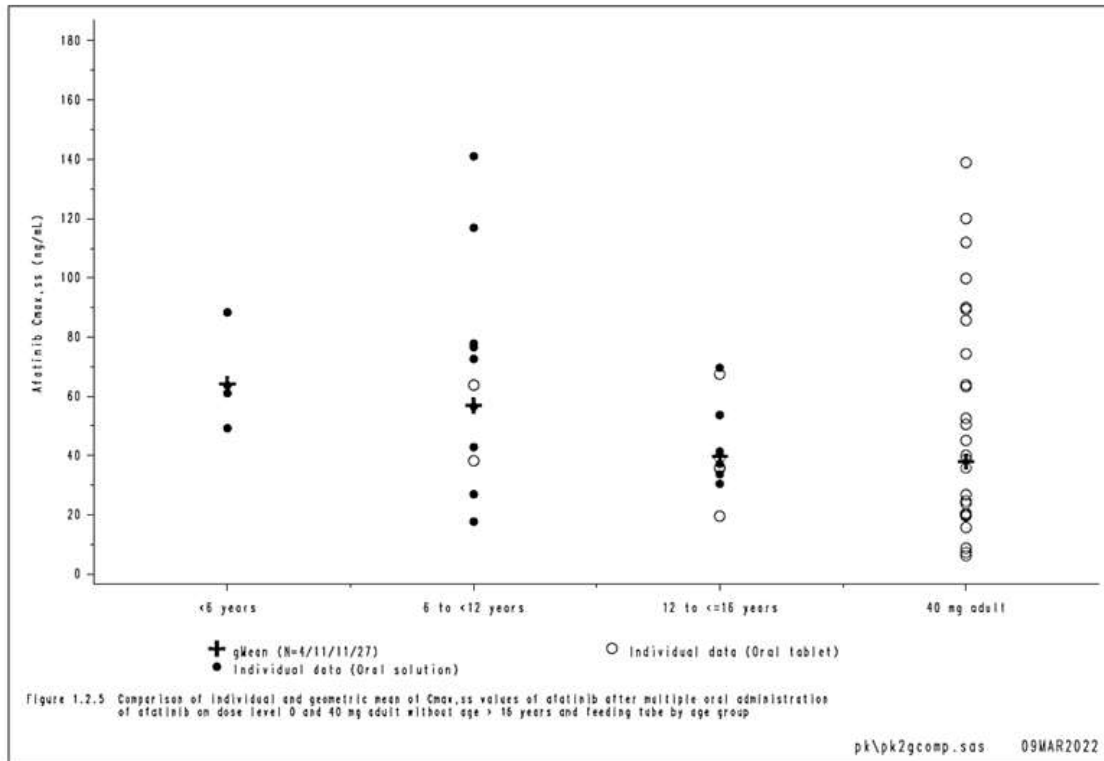


Figure 3. Comparison of individual and gMean of C_{max,ss} values of afatinib at 18mg/m² QD and adults at 40mg QD. (source: PopPK report - c31792800-01)



9 Appendices

Appendix 1 Analysis dataset description

Fifty-six patients 2 to ≤ 18 years of age were enrolled in the dose- finding part ($n=17$; 18mg/m²/day or 23 mg/m²/day) or expansion part ($n=39$; 18mg/m²/day). o Patients received afatinib as the commercially available tablets or unapproved solution, orally or using a feeding tube. Among the 56 patients, 33 and 16 patients were pediatric patients (<17 years old, $n=49$) who received afatinib in the form of solution or tablets, respectively. Because only 3 patients received afatinib solution using a feeding tube, clinically meaningful differences could not be evaluated between the estimated afatinib PK parameters from these patients and from those who received the commercial tablets. Therefore, these patients were omitted from the PK analysis. Nine of the 46 remaining pediatric patients received a higher dosage. Hence, only 37 pediatric patients (Table 1) who received a dosage of 18 mg/m²/day were included in the PK analysis to evaluate for any PK differences between pediatric patients and adults at the recommended dosage of 40 mg.

ZENSAR

CONFIDENTIAL

2034326535_00

Appendix 2 Comparison of afatinib PK by dosage form

Among the 56 patients enrolled into the study, 22, 31 and 3 received the approved commercial tablets, the unapproved solution (orally) or unapproved solution (via feeding tube), respectively